

$$1) a - P \begin{pmatrix} 2 & -2 & 1 & 2 \\ 1 & 1 & 1 & -1 \\ 1 & -1 & 1 & 1 \end{pmatrix}$$

$$P^T \begin{pmatrix} 2 & 1 & 1 \\ -2 & 1 & -1 \\ 1 & 1 & 1 \\ 2 & -1 & 1 \end{pmatrix}$$

$$P \cdot P^T = \begin{pmatrix} 2 & -2 & 1 & 2 \\ 1 & 1 & 1 & -1 \\ 1 & -1 & 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 2 & 1 & 1 \\ -2 & 1 & -1 \\ 1 & 1 & 1 \\ 2 & -1 & 1 \end{pmatrix}$$

$$= 2 + 2 + 1 + 2 = 7$$

$$b - \begin{pmatrix} 2 & -2 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix} \cdot \begin{pmatrix} a & 1/2 & -3/2 \\ 0 & b & -1/2 \\ -1 & 0 & c \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2a-1 & 1-2b & -2+c \\ a-1 & 1/2+b & -2+c \\ a-1 & 1/2-b & -1+c \end{pmatrix}$$

$$a = 1 \quad b = 1/2 \quad c = 2$$

$$a + b + c = 1 + \frac{1}{2} + 2 = \frac{2}{2} + \frac{1}{2} + \frac{4}{2} = \frac{7}{2}$$

$$c - (-1)^{3+2} \times \begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} = -1 \times 1 = -1$$

$$d - \pi \begin{pmatrix} 2 & -2 & 1 & 2 \\ 1 & 1 & 1 & -1 \\ 1 & -1 & 1 & 1 \end{pmatrix}$$

$$\xrightarrow[\substack{L_2 \leftarrow L_2 - L_3}]{OL_1} \begin{pmatrix} 2 & -2 & 1 & 2 \\ 0 & 2 & 0 & -2 \\ 1 & -1 & 1 & 1 \end{pmatrix} \xrightarrow[\substack{L_1 \leftarrow L_1 - L_3}]{OL_2} \begin{pmatrix} 1 & -1 & 0 & 1 \\ 0 & 2 & 0 & -2 \\ 1 & -1 & 1 & 1 \end{pmatrix}$$

$$\xrightarrow[\substack{L_3 \leftarrow L_3 - L_1}]{OL_3} \begin{pmatrix} 1 & -1 & 0 & 1 \\ 0 & 2 & 0 & -2 \\ 0 & 0 & 1 & 0 \end{pmatrix} \xrightarrow[\substack{L_2 \leftarrow L_2 \times \frac{1}{2}}]{OL_4} \begin{pmatrix} 1 & -1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$\xrightarrow[\substack{L_1 \leftarrow L_1 + L_2}]{OL_5} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

2) $b = 0$ e a pode ser qualquer valor

$$3) \begin{pmatrix} a & b & c \\ 1 & 1 & 1 \\ d & e & f \end{pmatrix} \xrightarrow[\substack{L_2 \leftarrow L_1}]{\substack{\lambda(-1)}} \begin{pmatrix} 1 & 1 & 1 \\ a & b & c \\ d & e & f \end{pmatrix} \xrightarrow[\substack{L_2 \leftarrow -L_2}]{\substack{\lambda(-1)}} \begin{pmatrix} 1 & 1 & 1 \\ -a & -b & -c \\ d & e & f \end{pmatrix}$$

$$\xrightarrow[\substack{L_2 \leftarrow L_2 + L_1}]{\substack{\lambda 2}} \begin{pmatrix} 1 & 1 & 1 \\ -a+1 & -b+1 & -c+1 \\ d & e & f \end{pmatrix} \xrightarrow[\substack{L_1 \leftarrow 2L_1}]{\substack{\lambda 2}} \begin{pmatrix} 2 & 2 & 2 \\ -a+1 & -b+1 & -c+1 \\ d & e & f \end{pmatrix}$$

$$\xrightarrow[\substack{L_2 \leftarrow L_2 + L_3}]{\substack{\lambda 2}} \begin{pmatrix} 2 & 2 & 2 \\ -a+1+d & -b+1+e & -c+1+f \\ a & b & c \end{pmatrix}$$

$$-3 \times (-1) \times (-1) \times 2 = 6$$

$$4) \begin{pmatrix} n & a & a & a \\ a & n & a & a \\ a & a & n & a \\ a & a & a & n \end{pmatrix} \xrightarrow[\substack{OL_1 \\ L_4 \leftarrow L_4 - L_3}]{} \begin{pmatrix} n & a & a & a \\ a & n & a & a \\ a & a & n & a \\ 0 & 0 & a-n & n-a \end{pmatrix}$$

$$\begin{cases} a-n=0 \\ n-a=0 \end{cases} \Leftrightarrow \begin{cases} a=n \\ n=a \end{cases}$$

Se tiver uma
linha nula,
 $\det = 0$

$$\boxed{n=a}$$

II

$$1) a - A^3 = A^2 \times A \\ = 8A^{-1} \times A = 8I \quad \checkmark$$

$$b - \det(A) \times \det(A) = 8^2 \times \frac{1}{\det(A)}$$

$$\Leftrightarrow \frac{(\det(A))^2}{\frac{1}{\det(A)}} = 64$$

$$\Leftrightarrow (\det(A))^3 = 64$$

$$\Leftrightarrow \det(A) = \sqrt[3]{64} = 4$$

$$c- \begin{vmatrix} 1 & 2 \\ 3 & 6 \end{vmatrix} = 6 - 6 = 0$$

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d-v

2) a-

$$3) a- (-1)^2 \times 0 \times \begin{vmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{vmatrix} + (-1)^3 \times 0 \times \begin{vmatrix} -1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix}$$

$$+ (-1)^4 \times 1 \times \begin{vmatrix} -1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix} + (-1)^5 \times \begin{vmatrix} -1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{vmatrix}$$

0

$$\begin{aligned} & -1 \times 0 \times 1 + 1 \times 1 \times 1 + 1 \times 0 \times 1 \\ & - 1 \times 0 \times 0 - (-1) \times 1 \times 1 - 1 \times 1 \times 1 \end{aligned}$$

$$= 1$$

$$b- \begin{pmatrix} 0 & 0 & 1 & 1 \\ -1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 & 1 & 1 \\ -1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 2 & 1 & 2 \\ 0 & 1 & 2 & 2 \\ 0 & 2 & 2 & 3 \end{pmatrix} \xrightarrow[\substack{OL1 \\ L_4 \leftarrow L_4 - 2L_3}]{OL1} \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 2 & 1 & 2 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & -2 & -1 \end{pmatrix}$$

$$\xrightarrow[\substack{OL2 \\ L_3 \leftarrow L_3 - \frac{1}{2}L_2}]{OL2} \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 2 & 1 & 2 \\ 0 & 0 & \frac{3}{4} & 1 \\ 0 & 0 & -2 & -1 \end{pmatrix} \xrightarrow{OL3} \textcircled{V}$$

$$C = \underbrace{\begin{pmatrix} 0 \\ -1 \\ 1 \\ 0 \end{pmatrix}}_{\substack{\text{first} \\ \text{column} \\ A'}} \cdot \begin{pmatrix} 0 & 0 & 1 & 1 \\ -1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix} \quad \textcircled{V}$$

$$d = \begin{pmatrix} 0 & 0 & 1 & 1 \\ -1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 0 & 1 & 1 \\ -1 & 0 & 1 & 1 \\ 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 0 & 1 & 1 & | & 0 \\ -1 & 0 & 1 & 1 & | & 0 \\ 1 & 0 & -1 & 1 & | & 0 \\ 0 & 1 & 1 & 0 & | & 0 \end{pmatrix} \xrightarrow[\substack{OL_1 \\ L_2 \leftarrow L_2 + L_3}]{\substack{OL_1 \\ L_2 \leftarrow L_2 + L_3}} \begin{pmatrix} -1 & 0 & 1 & 1 & | & 0 \\ 0 & 0 & 0 & 2 & | & 0 \\ 1 & 0 & -1 & 1 & | & 0 \\ 0 & 1 & 1 & 0 & | & 0 \end{pmatrix}$$

$$\xrightarrow[\substack{OL_2 \\ L_3 \leftarrow L_3 + L_1}]{\substack{OL_2 \\ L_3 \leftarrow L_3 + L_1}} \begin{pmatrix} -1 & 0 & 1 & 1 & | & 0 \\ 0 & 0 & 0 & 2 & | & 0 \\ 0 & 0 & 0 & 2 & | & 0 \\ 0 & 1 & 1 & 0 & | & 0 \end{pmatrix} \xrightarrow[\substack{OL_3 \\ L_4 \leftarrow L_2 \\ L_3 \leftarrow L_3 - L_4}]{\substack{OL_3 \\ L_4 \leftarrow L_2 \\ L_3 \leftarrow L_3 - L_4}} \begin{pmatrix} -1 & 0 & 1 & 1 & | & 0 \\ 0 & 1 & 1 & 0 & | & 0 \\ 0 & 0 & 0 & 2 & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \end{pmatrix}$$

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III

$$1) a - \begin{pmatrix} 1 & \alpha - 1 & \beta & | & -1 \\ 1 & 0 & \beta & | & \beta + 2 \\ 1 & \alpha & -3 & | & -1 \end{pmatrix} \xrightarrow[\substack{OL_1 \\ L_1 \leftarrow L_1 - L_2}]{\substack{OL_1 \\ L_1 \leftarrow L_1 - L_2}} \begin{pmatrix} 0 & \alpha - 1 & 0 & | & -\beta - 3 \\ 1 & 0 & \beta & | & \beta + 2 \\ 1 & \alpha & -3 & | & -1 \end{pmatrix}$$

$$\xrightarrow[\substack{OL_2 \\ L_3 \leftarrow L_3 - L_2}]{\substack{OL_2 \\ L_3 \leftarrow L_3 - L_2}} \begin{pmatrix} 0 & \alpha - 1 & 0 & | & -\beta - 3 \\ 1 & 0 & \beta & | & \beta + 2 \\ 0 & \alpha & -3 - \beta & | & -\beta - 3 \end{pmatrix} \xrightarrow[\substack{OL_3 \\ L_1 \leftarrow L_1 - L_3}]{\substack{OL_3 \\ L_1 \leftarrow L_1 - L_3}} \begin{pmatrix} 0 & \alpha - 1 & 0 & | & -\beta - 3 \\ 1 & 0 & \beta & | & \beta + 2 \\ 0 & \alpha & -3 - \beta & | & -\beta - 3 \end{pmatrix}$$